

Ans-1

$$K = \text{mod}(22101698, 245) + 1$$
$$= 3 + 1 = 4$$

$I =$

34	22	12	10	9
14	13	5	11	13
17	3	4	3	3
28	29	5	21	33
30	27	13	25	31

$$K_1 = \begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline 2 & 0 & -2 \\ \hline 1 & 0 & -1 \\ \hline \end{array}$$

$$K_2 = \begin{array}{|c|c|c|} \hline \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \hline \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \hline \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \hline \end{array}$$

(a)

$$K'_1 = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

$$(I \times K_1)_{11} = -34 + 0 + 12 - 28$$
$$+ 0 + 10 - 17 + 0 + 4$$
$$= -53$$

$$(I \times K_1)_{12} = -22 + 0 + 10 - 26$$
$$+ 0 + 22 - 3 + 0 + 3$$
$$= -16$$

$$\therefore I * K_1 =$$

-53	-16	12
-58	-10	34
-26	-18	23

(b)

$k_2' =$

$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$

$$(I \star k_2)_{11} = \frac{34}{9} + \frac{22}{9} + \frac{12}{9} + \frac{14}{9} + \frac{13}{9} + \frac{5}{9} + \frac{17}{9} + \frac{3}{9} + \frac{4}{9} = \frac{124}{9} = 13.78$$

$I \star k_2 =$

13.78	9.22	7.78
13.11	10.44	10.89
17.33	14.44	15.33

(c)

$k_1 =$

1	0	-1
2	0	-2
1	0	-1

$$(I \star k_1)_1 = 34 + 28 + 17 - 12 - 10 - 4 = 53$$

$$(I \star k_1)_2 = 22 + 26 + 3 - 10 - 22 - 3 = 16$$

$I \star k_1 =$

53	16	-12
58	10	-34
76	18	-73

(d)

$I \star k_2 =$

13.78	9.22	7.78
13.11	10.44	10.89
17.33	14.44	15.33

$$(I \star k_2)_{11} = \frac{34}{9} + \frac{22}{9} + \frac{12}{9}$$

$$+ \frac{14}{9} + \frac{13}{9} + \frac{5}{9} + \frac{17}{9} + \frac{3}{9} + \frac{4}{9} = \frac{124}{9} = 13.78$$

(e)

From (a) and (c) we can see, ~~$I \star k_1$~~

$$I * k_1 \neq I \star k_1$$

Also, k_1 is not symmetric; so the statement

$I * k_1 = I \star k_1$ is false.

From (b) and (d) we can see,

$$I * k_2 = I \star k_2$$

Also, k_2 is symmetric; so the statement

$I * k_2 = I \star k_2$ is true.

Ans-2

$$k = \text{mod}(22101698, 9) + 1 = 2 + 1 = 3$$

-11	-7	-4	-1	3	3	11	7	4	9	3	23
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(a)

$$S * k_1 =$$

11	-15	1	0	-1	4	-8	12	-1	-8	11	-26	43	-23
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$$0 + 0 + (-1 \times -11) = 11$$

$$0 + (2 \times -1) + (-7 \times -1) = -15$$

(b)

$$S * k_1 =$$

$$0 + 0 + (-1 \times -11) = 11$$

$$0 + (2 \times -1) + (-7 \times -1) = -15$$

11	-15	1	0	-1	4	-8	12	-1	-8	11	-26	43	-23
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(c)

$$S * k_2 =$$

-4	-1.2	2.4	4.6	5.6	6.8	6.8	9.2
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$$-\frac{11}{5} + \frac{-7}{5} + \frac{-4}{5} + \frac{-1}{5} + \frac{3}{5} = -4$$

$$-\frac{7}{5} - \frac{4}{5} - \frac{1}{5} + \frac{3}{5} + \frac{3}{5} = -\frac{6}{5} = -1.2$$

(d)

$$S \star k_2 =$$

-4	-1.2	2.4	4.6	5.6	6.8	6.8	9.2
----	------	-----	-----	-----	-----	-----	-----

$$-\frac{11}{5} - \frac{7}{5} - \frac{4}{5} - \frac{1}{5} + \frac{3}{5} = -4$$

$$-\frac{7}{5} - \frac{4}{5} - \frac{1}{5} + \frac{3}{5} + \frac{3}{5} = -\frac{6}{5} = -1.2$$

Ans - 3

$$k = \text{mod} (22101698, 7) + 3 = 6$$

$$\text{LPF} = \frac{1}{3} \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \hline \end{array}$$

↑

$$C1 = \begin{bmatrix} -11 & -7 & -4 & -1 & 3 & 6 & 11 & 7 & 4 & 9 & 7 & 23 \end{bmatrix}$$

$$-\frac{11}{3} - \frac{7}{3} - \frac{4}{3} = -7.33$$

$$\text{filtered } C1 = \begin{bmatrix} -7.33 & -4 & -0.67 & 2.67 & 6.67 & 8 & 7.33 & 6.67 & 13 \end{bmatrix}$$

$$C2 = \begin{bmatrix} -9 & -3 & -1 & 3 & 5 & 7 & 6 & 3 & 4 & 9 & 11 & 6 \end{bmatrix}$$

$$-\frac{9}{3} - \frac{3}{3} - \frac{1}{3} = -4.33$$

$$\text{filtered } C2 = \begin{bmatrix} -4.33 & -0.33 & 2.33 & 5 & 6 & 5.33 & 4.33 & 5.33 & 8 & 8.67 \end{bmatrix}$$

$$C3 = \begin{bmatrix} 7 & 9 & 15 & 17 & 21 & 13 & 11 & 6 & 4 & -3 & -13 & -7 \end{bmatrix}$$

$$\frac{7}{3} + \frac{9}{3} + \frac{15}{3} = 10.33$$

$$\text{filtered } C3 = \begin{bmatrix} 10.33 & 13.67 & 17.67 & 17 & 15 & 10 & 7 & 2.33 & -4 & -7.67 \end{bmatrix}$$

$$-7.33 - 4.33 + 10.33 = -1.33$$

filtered Output =

$$\begin{bmatrix} -1.33 & 9.34 & 19.33 & 24.67 & 27.67 & 23.33 & 18.63 & 14.33 & 10.67 \end{bmatrix}$$

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ReLU (filtered Output) =

$$\begin{bmatrix} 0 & 9.34 & 19.33 & 24.67 & 27.67 & 23.33 & 18.63 & 14.33 & 10.67 & 14 \end{bmatrix}$$

Ans 4

	Layer	Parameter count	Output variable	Dimensions
Image line	C_1^I	$7^2 * 3 * 64 + 64 = 9472$	I_1	[64, 64, 256, 256]
	M_1^I	0	I_2	[64, 64, 128, 128]
	C_2^I	$(2 * 7)^2 * 64 * 128 + 128 = 1605760$	I_3	[64, 128, 128, 128]
	M_2^I	0	I_4	[64, 128, 32, 32]
	G^I	0	I_5	[64, 128]
Sound line	C_1^S	$9 * 2 * 32 + 32 = 608$	S_1	[64, 32, 1024]
	M_1^S	0	S_2	[64, 32, 256]
	C_2^S	$9 * 32 * 64 + 64 = 18496$	S_3	[64, 64, 256]
	M_2^S	0	S_4	[64, 64, 64]
	G^S	0	S_5	[64, 64]
Combined line	L_1^C	$192 * 64 + 64 = 12352$	F_1	[64, 192]
	L_2^C	$64 * 1 + 1 = 65$	F_2	[64, 64]